

Machine Learning Challenge

The machine learning challenge is intended to be a practical and fun, cumulative project, to be completed in teams of 1-4 students.

We have collected data that asked students questions about several popular cities. Part of this data will be shared with you in a CSV file. Your task is to use the provided data to build a classifier related to this data set, to predict which city a student is referring to in their responses.

Your goal is to build a classifier that can perform well on an unseen test set that will *not* be shared with you. Your model is expected to perform reasonably on this unseen data.

You are welcome to use any materials you want from this course. You can choose any model you'd like, or use an *ensemble* of multiple models, as long as your model files do not exceed 10MB in size.

One important aspect of this challenge is that the test set is not shared with you, and that your model should perform reasonably on this data. To do well, be mindful of when your model underfits/overfits.

To make this fun, there will be a prize (to be determined) for the group(s) who perform the best on the unseen test set!

Machine Learning Challenge Components

The ML challenge is worth 14% of your final grade in this course. It has three components:

Data Collection (1%) – Due June 1, 11:59pm

Completion of the data collection, which will be in the form of a Quercus quiz with several questions about 4 popular cities. This should take 5-10 minutes to complete.

Model Predictions(4%) – Due July 31, 10pm

Submit a python3 script called `pred.py` that has a function `predict_all` that takes, as a parameter, the name of a CSV file containing the test set, and predictions for that test set. Your script can use the following imports:

- `numpy`
- `pandas`
- basic python imports: `sys`, `csv`, `random`

We will be using python3.10 on the Markus system.

You will not be able to use `sklearn`, `pytorch`, `tensorflow`, etc. in the final submitted code, so that you have a chance to build models yourselves. You may reuse any code that you wrote (or was provided to you) in any of the labs. However, during the exploration phase of your project, you are welcome to use whatever tools you want before deciding what model you would like to build.

Note that your code only needs to be able to make predictions. You can still use any code/data you want to *build* those models!

You can submit additional files that are used by `pred.py`, e.g., to store your model parameters. All of your files, taken together, should be at most 10MB in size. Your script should not require networking, and should not download any new files. Your model script should be reasonable in its use of memory resources, and should be able to make ~60 predictions within 1 minute.

Your model will be graded for the following:

- Model script is runnable, meets the criteria for file size, imports, etc., and produces reasonable results over the test set. We will set the “reasonable” threshold such that groups who follow good machine learning practises and choose reasonable models should be able to pass the threshold.

PDF Report (9%) – Due July 31, 10pm

Submit a pdf file called `csc311challenge.pdf` that describes your final model, as well as the steps that you took to develop this model.

Along with the PDF file, please also submit any other `.py` or `.ipynb` files that you used while developing your model. This latter files will not be graded, but will be used as evidence that you completed your own work. The `.py` or

.ipynb files do not need to be runnable by the TA, and can rely on external imports (e.g., sklearn) and data sets that you don't submit. Only your final `pred.py` file needs to be runnable.

Your report should describe:

- **Data:** Describe how you explored your data, and the process through which you determined your input features. How did you end up representing your data? What else did you try? We are looking for:
 - A thorough exploration of the data, similar to what is presented in labs. What are the distributions of the features? How do these features correlate with the target?
 - If you use figures to answer those questions, an explanation of how you are interpreting those figures. Take care that your figures can be interpreted meaningfully.
 - A clear and logical determination of how you determined your input features, with convincing logical or empirical evidence justifying your choice. Important features should not be overlooked (e.g., not removed for “ease”).
 - A clear description of the way(s) that you are representing the data in your models.
 - The descriptions should be consistent with the `.py` and/or `.ipynb` files that you used while developing your model.
 - A clear description of how you are splitting your data into various sets. You may use k-fold cross validation if you'd like, but if you do, describe how you are applying that idea.
- **Model:** A description of the model(s) that you are evaluating/exploring. We are expecting a thorough exploration of at least 3+ families of models, even if you don't ultimately use those models. We are looking for:
 - How you are applying these models. You don't need to reiterate what the models are and how they work. Instead, we're looking for a description of the choices you are making to apply the model to this task.
- **Model Choice and Hyperparameters:** How you are determining what model(s) to use in your final `pred.py`, as well as an exploration of hyperparameters. We are looking for:
 - A convincing explanation of how you ensured that the evaluation metrics for various models are comparable (i.e., are you using a consistent test set?)
 - A clear description of the evaluation metric(s) used to evaluate your model, as well as justification for its use.
 - A description of the hyperparameters that you are tuning, hyperparameter combinations that you have tried, and the corresponding evaluation metric(s). We are not looking for an exhaustive search of all possible hyperparameter combinations, but there should be enough evidence to demonstrate that your hyperparameter choices are reasonable.
 - A clear description of what your final model choice looks like in the submitted `pred.py` file.
 - The descriptions should be consistent with the `.py` and/or `.ipynb` files that you used while developing your model.
- **Prediction:** How well do you expect your model to perform on the test set? We are looking for:
 - A point estimate for your performance, not a range.
 - A reasoned explanation of your expected model performance, with empirical evidence supporting your explanation. You are not graded on the closeness of your estimate to the final test accuracy, but you are graded on your reasoning.
- **Workload Distribution:** A description of what each person in the group contributed to the project. A 1-2 sentence description of each person's role is sufficient. Each student's description must be written by that student in order for them to receive credit for the project.

Recommendations

Start early! The data exploration portion can be done as soon as data is available.

Communicate early and often. There are a lot of tasks to do as a team and decisions to be made.

Keep a journal (e.g., google doc or overleaf) and record the experiments that you perform. If you keep clear, reproducible records, then writing the **Model Choice and Hyperparameter** section of your report will be much easier.

Try out various models using sklearn before writing your own implementation. Leverage the code that you have already written for the labs.

Be mindful that machine learning models can take a while to train, and that some models are more time-consuming to train than others.